

Validation Memo

Calculation reconciled; general frequency superiority unsupported

Scope and disposition

Accept as a reproducible, bounded historical validation case. Compare nine sector ETFs across two signals, six semiannual schedules and four costs, plus six-sleeve pooled accounts. Primary inference is 2001-2025, 12-1, 5 bps. Do not use this case to certify a live strategy or choose an optimal calendar.

Data lineage

Preserved Yahoo responses -> common-session adjusted prices -> portfolio accounts -> accounting replay -> tranche sums -> monthly risk and paired intervals -> public derived CSVs and summary. SHA-256 binds snapshots, configuration and code. French monthly RF is captured separately and joined to 300 months. Hash identity does not certify source accuracy.

Separate implementation checks

Legacy replay: 60 paths. Mechanism replay: 108 paths, 726,732 daily states, maximum relative NAV error $1.78e-14$. Extension: eight tranches, 53,832 daily states and 34,800 monthly rows; capital, cash, units, gross fees, risk and bootstrap percentiles reconcile. The extension verifier imports no project calculation code.

Independence boundary

The same AI agent authored both implementations. Separate algorithms can expose arithmetic and implementation defects but cannot exclude a shared misunderstanding of the specification. No external reviewer agreement, independent price validation or live fill certification is claimed.

Findings and residual risk

The primary difference changes sign across calendars; both paired intervals span zero. Tranching retains a deeper drawdown. Prior history exposure, adjusted-price/opening proxies, fixed costs, zero-interest cash, changing sector exposures and bootstrap nonstationarity remain risks. Written owner result and judgment reviews are accepted; signal-window explanation and observed oral presentation remain pending.

Evidence: Study and downloads | Repository, methods and receipts

Method/source: French Data Library monthly RF; Politis and Romano (1994), The Stationary Bootstrap. Exact URLs and capture hashes appear in the reproduction guide.